

- This exam contains 12 pages (including the front and back of this cover page) and 14 problems. Check to see if any pages are missing.
- This is the second exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 2 bonus questions worth a total of 6 possible points.

Page:	3	4	5	6	7	8	9	10	Total
Points:	20	10	10	12	10	10	10	18	100
Score:									

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

For readability in the key, correct answers are colored in red, e.g. **Correct Answer**, while incorrect answers are marked with a strike, e.g. ~~Incorrect Answer~~.

2 pts

- (a) If $f(x) = 3(5)^{x-2} + 2$, then $f(7) = 9377$.

☒ True☐ False

2 pts

- (b) The logarithmic equation $\log_7(142) = a$ is equivalent to the exponential equation $7^a = 142$.

☒ True☐ False

2 pts

- (c) The exponential equation $e^x = y$ is equivalent to the logarithmic equation $\ln(x) = y$.

☐ True☒ False

2 pts

- (d) $\log_\pi(\sqrt{x^5 y^{-10}}) = 5/2 \log_\pi(x) - 5 \log_\pi(y)$.

☒ True☐ False

2 pts

- (e) The angle $5\pi/4$ in radians is equivalent to the angle 225° in degrees

☒ True☐ False

2 pts

- (f) The angle between 0 and 2π in radians that is coterminal to the angle $-3\pi/4$ is $5\pi/4$.

☒ True☐ False

2 pts

- (g) $\sin(\pi/3) = \sqrt{2}/2$.

☐ True☒ False

2 pts

- (h) $\sec\left(\frac{\pi}{3}\right) = 2$

☒ True☐ False

2 pts

- (i) If $\sec(x) = 4.5$, then $\sec(-x) = -4.5$.

☐ True☒ False

2 pts

- (j) $\csc(x)$ is the inverse function of $\sin(x)$.

☐ True☒ False

5 pts

2. Use the definition of a logarithm to solve the equation

$$12 - 2\log_{10}(8x + 4) = 8.$$

☐ (A) $x = 3$

☐ (B) $x = e^4$

☒ (C) $x = 12$

☐ (D) $x = 0$

$$\begin{aligned}12 - 2\log_{10}(8x + 4) &= 8 \\-2\log_{10}(8x + 4) &= -4 \\\log_{10}(8x + 4) &= 2 \\10^{\log_{10}(8x+4)} &= 10^2 \\8x + 4 &= 100 \\8x &= 96 \\x &= 12\end{aligned}$$

5 pts

3. Solve the equation for x :

$$\ln(7) - \ln(7 - 7x) = \ln(8).$$

☒ (A) $x = 7/8$

☐ (B) $x = e^8$

☐ (C) $x = -7/8$

☐ (D) $x = 12$

$$\begin{aligned}\ln(7) - \ln(7 - 7x) &= \ln(8) \\\ln\left(\frac{7}{7 - 7x}\right) &= \ln(8) \\\ln\left(\frac{1}{1 - x}\right) &= \ln(8) \\e^{\ln(\frac{1}{1-x})} &= e^{\ln(8)} \\\frac{1}{1 - x} &= 8 \\1 &= 8 - 8x \\8x &= 7 \\x &= 7/8.\end{aligned}$$

5 pts

4. Find the length of the arc of a circle of radius 10 inches subtended by a central angle of 50° .

(A) $25/9$ inches

(B) π inches

(C) $25/9\pi$ **inches**

(D) $9/25\pi$ inches

In a circle of radius r , the length, s , of an arc subtended by an angle with measure θ in radians is $s = r\theta$. So, we first convert 50° to radians:

$$(50^\circ) \left(\frac{\pi \text{ radians}}{180^\circ} \right) = 5\pi/18 \text{ radians.}$$

Using this, we reach our answer with the following computation:

$$s = (10 \text{ inches}) (5\pi/18) = (25/9) \pi \text{ inches}$$

5 pts

5. Find the area of a sector of a circle with radius 8 inches subtended by a central angle of $\pi/2$ radians.

(A) 2π inches²

(B) 32π inches²

(C) 8π inches²

(D) 16π **inches²**

The area of a sector of a circle with radius r subtended by an angle θ , measured in radians, is $A = 1/2\theta r^2$. So,

$$A = 1/2 (\pi/2) (8 \text{ inches})^2 = 16\pi \text{ inches}^2.$$

7 pts

6. Find the (x, y) -coordinates of the point on a circle with radius 25 corresponding to an angle of $\frac{\pi}{6}$.

Ⓐ $\left(\frac{25}{2}, \frac{25\sqrt{3}}{2}\right)$ Ⓑ $\left(\frac{25\sqrt{3}}{2}, \frac{25}{2}\right)$ Ⓒ $(25\sqrt{3}, 25)$ Ⓓ $\left(\frac{25\sqrt{2}}{2}, \frac{25\sqrt{2}}{2}\right)$

Since the (x, y) -coordinates of a point on a circle with radius r corresponding to an angle of θ radians is $(r \cos(\theta), r \sin(\theta))$, we reach our answer with the following computation:

$$x = (25) \cos\left(\frac{\pi}{6}\right) = \frac{25\sqrt{3}}{2}$$

$$y = (25) \sin\left(\frac{\pi}{6}\right) = \frac{25}{2}$$

5 pts

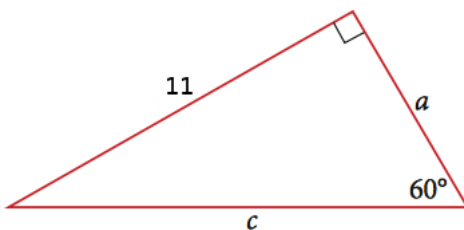
7. If $x \sin(x) = -4.5$, find $(-2x) \sin(-x)$.

Since $\sin$ is an odd function, we know that $\sin(-x) = -\sin(x)$. So, we find our answer by using this fact and computing as follows:

$$\begin{aligned} (-2x) \sin(-x) &= (-2x)(-\sin(x)) \\ &= (2x) \sin(x) \\ &= 2(x \sin(x)) \\ &= 2(-4.5) \\ &= -9. \end{aligned}$$

10 pts

8. Solve the unknown sides of the triangle below:



Ⓐ $a = \frac{11\sqrt{3}}{3}, c = \frac{22\sqrt{3}}{3}$

Ⓒ $a = \frac{11\sqrt{2}}{2}, c = \frac{11\sqrt{2}}{2}$

Ⓑ $a = 11, c = 22$

Ⓓ $a = 11, c = \frac{22\sqrt{3}}{3}$

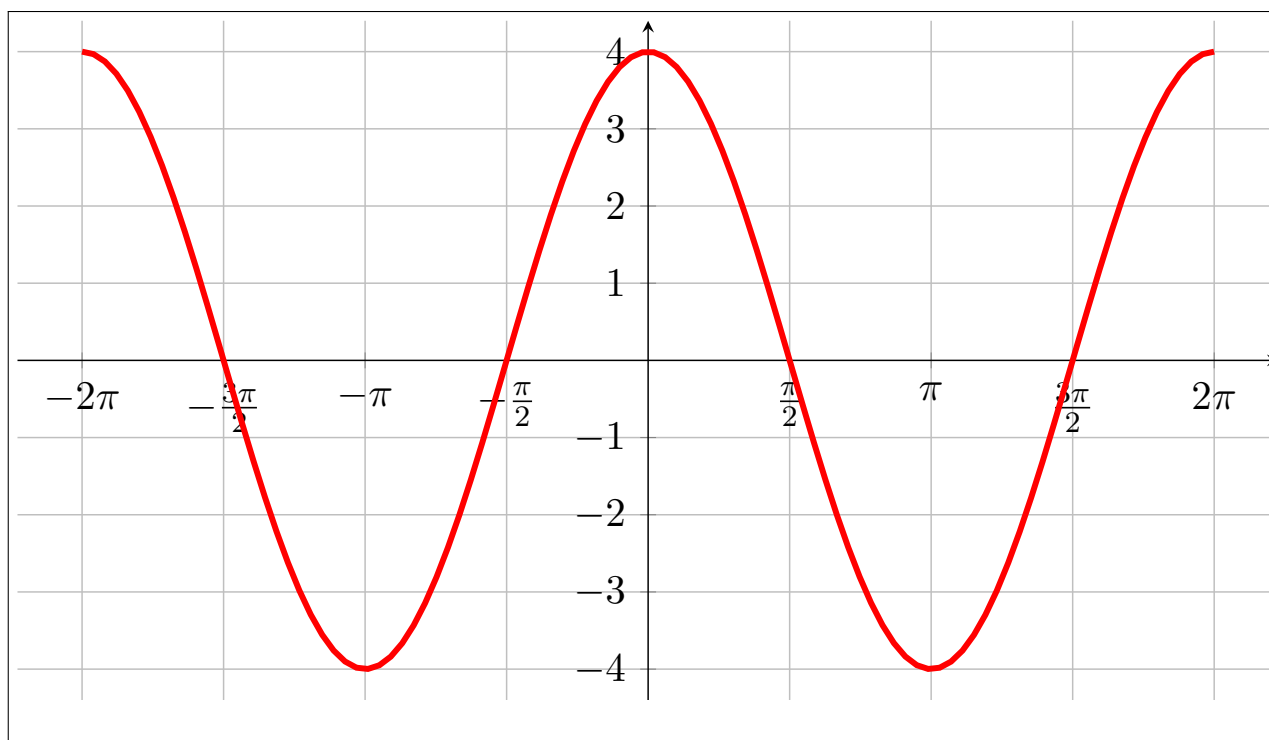
First, we solve for a using the tangent function:

$$\begin{aligned}\tan(60^\circ) &= \frac{11}{a} \\ a &= \frac{11}{\tan(60^\circ)} \\ a &= \frac{11\sqrt{3}}{3}.\end{aligned}$$

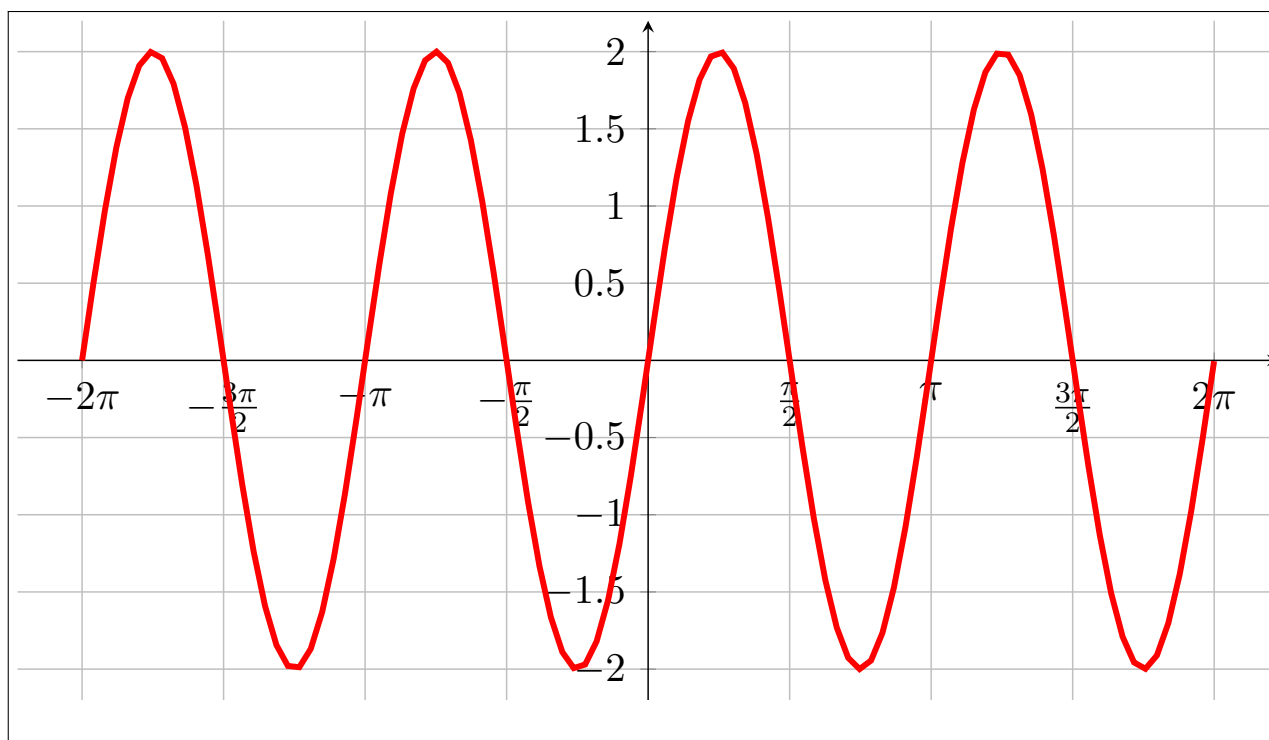
Then, we solve for c using the pythagorean theorem:

$$\begin{aligned}c &= \sqrt{a^2 + 11^2} \\ &= \sqrt{\left(\frac{11\sqrt{3}}{3}\right)^2 + 11^2} \\ &= \frac{22\sqrt{3}}{3}.\end{aligned}$$

10 pts

9. Graph the function $f(x) = 4 \cos(x)$.

- 10 pts 10. Graph the function $f(x) = 2 \sin(2x)$.



8 pts

11. State the amplitude, period, phase shift and vertical translation of the function

$$f(x) = 3 \sin \left(\frac{1}{3}x + \frac{3\pi}{4} \right) - 2$$

Amplitude : $A = 3$ Period : $P = \frac{2\pi}{B} = \frac{2\pi}{(1/3)} = 6\pi$ Phase Shift : $\frac{C}{B} = \frac{(-3\pi/4)}{(1/3)} = -\frac{9\pi}{4}$ Vertical Translation : $D = -2$

10 pts

12. Find all values of x in the interval $[0, 2\pi)$ such that $\cos(x) = -\frac{1}{2}$.☒ A $x = 2\pi/3$ and $x = 4\pi/3$ ☐ C $x = \pi/3$ and $x = 5\pi/3$ ☐ B $x = \sqrt{2}/2$ and $x = -\sqrt{2}/2$ ☐ D $x = 8\pi/3$ and $x = 10\pi/3$

To find the first solution, we simply look at the top half of the unit circle and notice that:

$$\cos(2\pi/3) = -1/2.$$

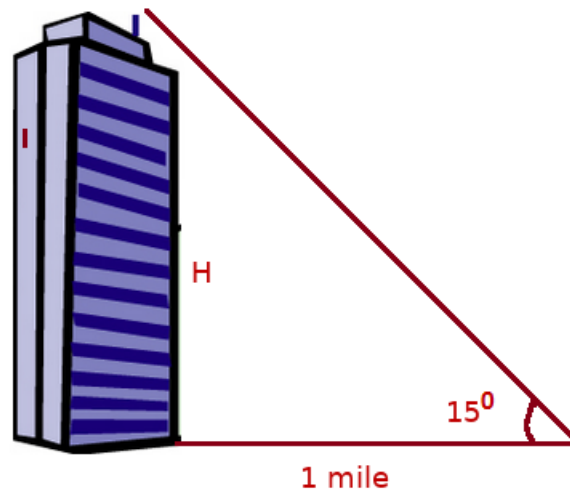
To find the second solution, we note that cosine is an even function. So, since $\cos(-x) = \cos(x)$, our second solution is $2\pi - 2\pi/3 = 4\pi/3$. Combining these results, we reach our answer:

$$x = 2\pi/3 \text{ and } x = 4\pi/3.$$

3 (bonus)

13. The angle of elevation to the top of a building is found to be 15° from the ground at a distance of 1 mile from the base of the building. Find the height of the building to the nearest foot. (Hint: 1 mile is 5280 feet)

Drawing a diagram, we have the following situation:



So, we first convert 15° to $(\pi/180^\circ)(15^\circ) = \pi/12$ radians. With this in hand, we can use the fact that

$$\tan(15^\circ) = \frac{H \text{ feet}}{5280 \text{ feet}}.$$

Doing some minor algebraic manipulation and using a calculator, we find that

$$H = \tan(\pi/12)(5280 \text{ feet}) \approx 1415 \text{ feet}.$$

3 (bonus)

14. The London Eye is a huge Ferris wheel with a diameter of 135 meters (443 feet). It completes one rotation every 30 minutes. Riders board from a platform 2 meters above the ground. Express a rider's height above ground as a function of time in minutes.

With a diameter of 135 meters, the wheel has a radius of 67.5 meters. The height will oscillate with amplitude 67.5 meters above and below the center. That is, $A = 67.5$.

Passengers board 2 meters above ground level, so the center of the wheel must be located $67.5 + 2 = 69.5$ meters above ground level. Thus, we can conclude that the midline of the oscillation will be at 69.5 meters. That is, $D = 69.5$.

The wheel takes 30 minutes to complete 1 revolution, so the height will oscillate with a period of 30 minutes. Since $P = 2\pi/B$, this yields that $30 = 2\pi/B$. Solving, this tells us that $B = 2\pi/30 = \pi/15$.

Lastly, because the rider boards at the lowest point, the height will start at the smallest value and increase.

So, the riders height h (in meters) as function of t (in minutes) is given by any one of the following equivalent answers:

$$\begin{aligned}h(t) &= -67.5 \cos(\pi/15t) + 69.5 \\&= -67.5 \sin(\pi/2 - \pi/15t) + 69.5 \\&= 67.5 \sin(\pi/15t - \pi/2) + 69.5\end{aligned}$$